

Metal-Decoration-Free Li_3C_2 Monolayer with Heptacoordinate Carbons as a Promising Hydrogen Storage Medium

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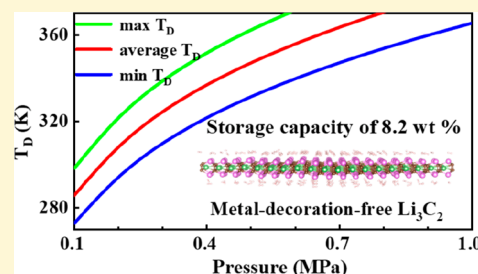


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ABSTRACT: Two-dimensional (2D) materials have been intensively studied as potential hydrogen storage mediums, because of the chemical stability, large specific surface area, and light weight. Here, a combination of crystal structure prediction and first-principles calculations predicts a metal-decoration-free Li_3C_2 monolayer that exhibits a hydrogen storage capacity of 8.2 wt % and high average hydrogen desorption temperature of 286 K. Li_3C_2 monolayer is constructed by two hexagonal Li layers sandwiched with Li–C layers containing a C dimer. Interestingly, each C atom binds with six neighboring Li atoms and one C atom, forming a quasi-planar heptacoordinate carbon moiety. Li_3C_2 monolayer can adsorb up to 2 H_2 molecules with the average adsorption energy of 0.22 eV/ H_2 , suitable for reversible hydrogen storage. Analysis suggests that the hybridization of Li s and H s orbitals plays a key role in hydrogen adsorption. Moreover, the exploration of the effect of temperature and pressure on hydrogen storage performance of Li_3C_2 monolayer shows that, at room temperature, all adsorbed hydrogen structures are stable at pressure of 40 bar. The calculated van't Hoff desorption temperatures and molecular dynamics simulations demonstrate the realization of hydrogen desorption at near room temperature and the effectiveness of hydrogen storage/release cycles. In addition, we provide a possible route to experimentally grow Li_3C_2 monolayer on suitable substrates, such as the Cu(111) surface and graphene.



Hydrogen, as a large-energy-density, abundant, clean, and high-conversion-efficiency energy source, can be utilized to be an efficient energy carrier, and it has been considered to be one of the most application prospects of energy.^{1,2} However, gaseous hydrogen storage and transport on a large scale are difficult, because of the low ignition energy and wide range of flammability. High-pressure gaseous hydrogen storage techniques possess low cost, convenience, and fast charging and discharging speeds; however, this necessitates advanced safety precautions.³ For liquid hydrogen storage, extra energy is necessary to liquefy and keep the hydrogen at cryogenic temperature, after which stage another problem may arise, known as boiling. As a contrast, solid-state hydrogen storage is a reversible and secure technique, i.e., hydrogen is adsorbed physically or chemically on the surface of a material, which could be the primary way for hydrogen storage in the future. The U.S. Department of Energy (DOE) set targets for ideal materials for hydrogen storage practical applications, namely, the hydrogen storage capacity is 5.5 wt % for onboard vehicle hydrogen storage systems in 2025 and the available conditions of 2–100 bar and –40–85 °C in operating vehicles.^{4,5} Moreover, the optimal hydrogen

adsorption energy should be in the range of 0.2–0.6 eV/ H_2 ,⁶ which is suited for reversible hydrogen storage under near standard conditions.

Carbon nanostructures^{7–12} and other common two-dimensional (2D) materials^{13–16} have been widely investigated as promising hydrogen storage mediums, because of the chemical stability, large specific surface area, and light weight, such as graphene,⁷ carbon nanotubes (CNTs),⁹ fullerenes,¹¹ silicene,¹³ BCN,¹⁴ MoS_2 ,¹⁵ $h\text{-BN}$,¹⁶ etc. Unfortunately, these intrinsic 2D materials have two fatal drawbacks, i.e., a hydrogen storage capacity of <5.5 wt % and a hydrogen desorption temperature that is far less than room temperature, which is due to the weak van der Waals interaction between hydrogen and the surface of a material.

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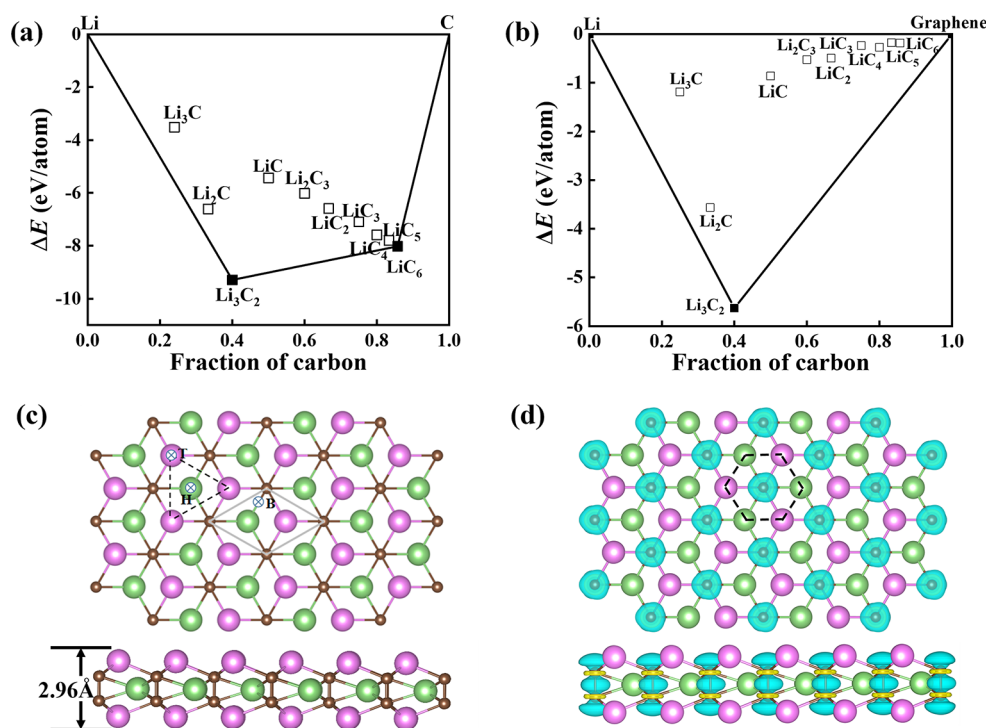


Figure 1. Convex hull diagrams of the Li–C system with respect to the decomposition into (a) isolated Li and C atoms and (b) the graphene and isolated Li atoms. Stable and metastable phases are marked by solid and hollow blocks, respectively. (c) Top and side views of Li_3C_2 monolayer with unit cell marked by gray rhombus frame. The pink, green, and brown spheres are surface-layer Li, middle-layer Li, and middle-layer C atoms, respectively. The possible adsorption sites of hydrogen are represented by H, B, and T, which are above the center of trigon, above the midline of Li–Li atoms, and top Li atoms, respectively. (d) Deformation charge density of Li_3C_2 monolayer. The yellow and blue refer to the charge depletion and accumulation regions, respectively. The isovalue is 0.008 e/a.u.

Metal decoration has been proved to be an effective method to improve the hydrogen storage capacity and resolve low-temperature effect of intrinsic 2D materials.^{11,17–33} In this case, hydrogen is adsorbed around the metal atom by the hybridization of orbitals³⁴ or polarization mechanism;²⁴ thus, there is a relatively strong interaction between metal and hydrogen, leading to larger storage capacity and high desorption temperature. For example, the hydrogen storage capacity of graphene has been increased to 6.0–18 wt % from 0.4 wt % after Li/Na/K/Ca/Al decoration.^{7,25,27} Hf_2CF_2 MXene layer after Li decoration has a hydrogen desorption temperature close to room temperature.²² Other examples include Ca-decorated zigzag graphene nanoribbon,²⁰ BC_2N ¹⁷ and BC_7 ,²⁸ graphdiyne¹⁹ and boron sheets,³⁴ K-decorated germanene,¹⁸ Mg-decorated silicene,²³ Li- and Na-codecorated boron phosphide,³⁵ Li-decorated BC_3 ²⁹ and BH system,³⁶ Li/Na/K decorated BN_2 ,³⁰ etc., which have storage capacities of 4.56–17.95 wt % and average desorption temperatures of 95–419 K. Nevertheless, with a high coverage of metal atoms, hydrogen adsorption is always weakened by the strong electrostatic interaction between Li cations. In addition, it is inevitable that decorated metals are easy to aggregate, and it is difficult to prepare experimentally for metal surface modification. Therefore, it is necessary to design an intrinsic 2D material with a storage capacity of >5.5 wt % and a desorption temperature close to room temperature.

In this work, we predict a thermodynamically stable Li_3C_2 monolayer with heptacoordinate carbons using a joint of crystal structure prediction and first-principles calculations. Li_3C_2 monolayer possesses a hydrogen storage capacity of 8.2 wt %, exceeding the minimum requirement set for onboard

vehicle hydrogen storage systems. The average adsorption energy of Li_3C_2 monolayer is 0.22 eV/ H_2 which is in the optimal hydrogen adsorption energy range. Moreover, $2\text{H}_2@ \text{Li}_3\text{C}_2$ exhibits the average hydrogen desorption temperature of 286 K, based on the van't Hoff equation, also confirmed by molecular dynamics simulations. These appealing properties render the Li_3C_2 monolayer a promising hydrogen storage medium for onboard vehicle applications.

Structure searches for 2D Li_xC_y ($x = 1, y = 1–6$; $x = 3, y = 1, 2$; $x = 2, y = 1$) system have been performed with simulation cells of 1–4 formula units (f.u.), using the swarm-intelligence based CALYPSO method.^{37–39} This method requires only chemical compositions for given compounds to predict stable or metastable structures, and has been an effective and popular tool to design new materials including 2D functional materials,^{40–43} superhard materials,^{44,45} high-energy-density materials,^{44,46} and high-temperature superconductors.^{47,48} The local structural relaxations and electronic properties calculations are based on the density functional theory (DFT),⁴⁹ as implemented in the Vienna ab initio simulation package (VASP).⁵⁰ The core electrons are treated by the projector augmented-wave approximation,⁵¹ and the exchange-correlation functional is given by the generalized gradient approximation parametrized by Perdew, Burke, and Ernzerhof.⁵² The plane-wave cutoff energy is set to be 700 eV in all calculations, and the first Brillouin zone is sampled using a $2\pi \times 0.03 \text{ \AA}^{-1}$ Monkhorst–Pack grid spacing.⁵³ All the geometries were fully relaxed until residual forces are smaller than $10^{-3} \text{ eV/\AA}$. The dynamic stabilities is identified from phonon calculations using the direct supercell method, as implemented in PHONOPY code.⁵⁴ Thermal stability was confirmed by ab

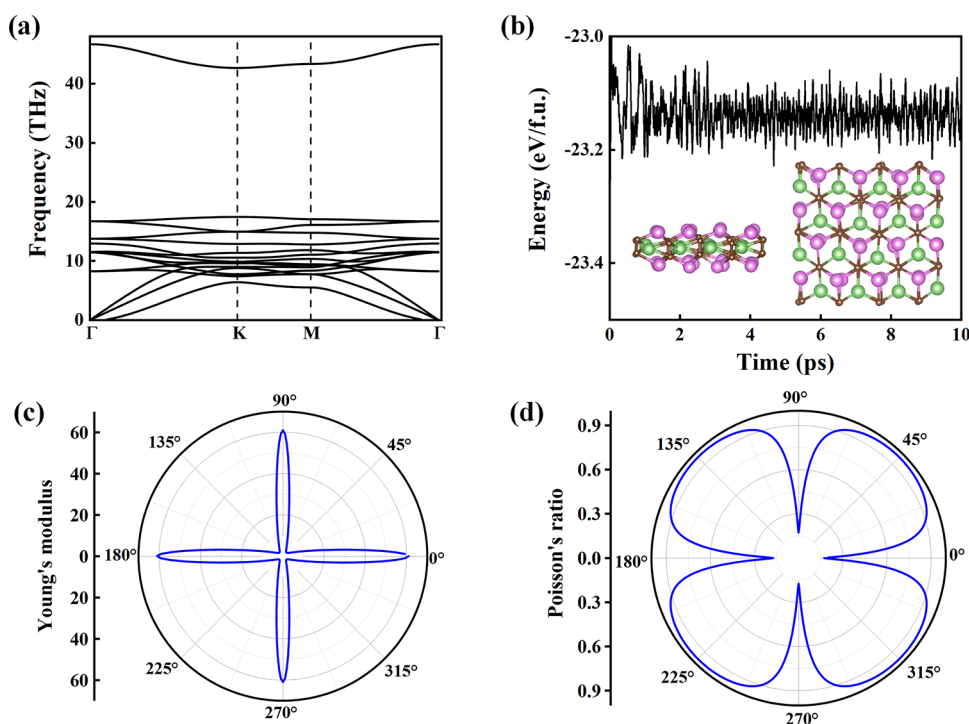


Figure 2. (a) Phonon dispersion curves of Li_3C_2 monolayer. (b) Evolution of energies during MD simulations of Li_3C_2 monolayer at 400 K. The inset is the snapshot of structure at the end of 10 ps simulations. Polar diagrams for the (c) Young's modulus and (d) Poisson's ratio of the Li_3C_2 monolayer.

initio molecular dynamics (AIMD) simulations,⁵⁵ which were performed with a time step of 1 fs, and a total simulation time of 10 ps. The temperature is maintained at 300, 400, and 500 K using a Nosé–Hoover chain thermostat. Sufficient vacuum thickness of 30 Å is used to avoid interaction of periodic images. The van der Waals (DFT-D2 method)⁵⁶ interactions are taken into account to accurately obtain hydrogen adsorption energy. The electron–phonon coupling (EPC) is calculated within linear-response theory using the QUANTUM ESPRESSO code.⁵⁷ Ultrasoft pseudopotentials for Li and C are used with a kinetic cutoff energy of 90 Ry.⁵⁸ $5 \times 5 \times 1$ q -point meshes (5 q -points) in the first Brillouin zone are used in the EPC calculation for Li_3C_2 monolayer. Correspondingly, a Monkhorst–Pack grid of $20 \times 20 \times 1$ is used to ensure k -point sampling convergence with Gaussians of width 0.02 Ry.

Figure 1a summarizes the formation energies (convex hull diagrams) of all considered Li–C compositions, with respect to the decomposition into isolated Li and C atoms, where points lying on the tie line correspond to the thermodynamically stable phases. Author: Is the last sentence correct as composed?? Its meaning is somewhat vague to me. Please advise. In addition to the known LiC_6 monolayer,⁵⁹ our unbiased structural searches also identify a new thermodynamically stable Li–C compounds with stoichiometry Li_3C_2 , which forms a vertex of the convex hull. The calculated formation energy of Li_3C_2 monolayer is -9.29 eV/atom, comparable with graphene (-10.13 eV/atom),⁶⁰ and lower than those of borophene (-5.99 eV/atom),⁶¹ silicene (-4.57 eV/atom),⁶² MoS_2 (-6.25 eV/atom)⁶³ and $h\text{-BN}$ (-7.91 eV/atom).⁶⁴ We also choose graphene as the reference, since it has been synthesized in experiment. As shown in Figure 1b, Li_3C_2 monolayer is still thermodynamically stable, indicating that incorporation of Li atoms into graphene is a possible route for synthesis.

Figure 1c presents the optimized crystal structure of Li_3C_2 monolayer, which adopts a hexagonal structure in $P\bar{6}m2$ symmetry with the lattice constant being $a = 3.553$ Å and $c = 38.598$ Å. In the Li_3C_2 monolayer, Li occupies (0, 0, 0.462) and (0.667, 0.333, 0.5) positions, as well as C occupies (0.333, 0.667, 0.482) position. As given in the side view, Li_3C_2 monolayer has a sandwiched feature of thickness of 2.96 Å, with two terminated Li layers and a middle Li–C layers containing C dimer. Interestingly, each C atom binds with neighboring six Li atoms (three terminated and three middle Li atoms) and one C atom, forming a quasi-planar heptacoordinate carbon (phC) moiety. The average bond length of Li–C is 2.21 Å, and the bond length of C–C is 1.32 Å. Deformation charge density is obtained by subtracting charge densities of isolated atoms from that of Li_3C_2 monolayer, which is used to understand the unique bonding of phC. As shown in Figure 1d, some electrons are extracted from the 2s orbitals of Li atoms and well-delocalized over the C–C and Li–C bonds, indicating that C atoms form multicenter covalent bonds with neighboring Li and C atoms, which is crucial for stabilizing the phC moieties. Moreover, we can find that some electrons the p orbitals of the C atoms back-donate Li atoms to further stabilize the phC moieties. The stabilization mechanism has been revealed in other 2D materials with a quasi-planar hypercoordinate motif, such as CaSi ⁶⁵ and Be_5C_2 .⁶⁶

The calculated phonon dispersion shows an absence of any imaginary frequencies in the whole Brillouin zone, confirming the dynamical stability of Li_3C_2 monolayer (Figure 2a). The thermal stability of Li_3C_2 monolayer is explored through AIMD simulations. As shown in Figure 2b, heating to 400 K for 10 ps, which revealed no structural destruction, indicates that Li_3C_2 monolayer is stable under ambient conditions. The elastic constants C_{ij} of Li_3C_2 monolayer have been calculated,

i.e., $C_{11} = 62.8$ N/m, $C_{12} = 10.89$ N/m, and $C_{66} = 25.98$ N/m, which meet the Born criteria for 2D materials⁶⁷ of $C_{11}C_{22} - C_{12}^2 > 0$ and $C_{66} > 0$, implying its mechanical stability. Based on the elastic constants, the orientation-dependent Young's modulus $E(\alpha)$ and Poisson's ratio $\nu(\alpha)$ ^{68–71} (α is the angle relative to the positive x direction) of Li_3C_2 monolayer have been plotted in Figures 2c and 2d. The polar diagrams show that $E(\alpha)$ and $\nu(\alpha)$ of Li_3C_2 monolayer are spatially varying. The maximum value of $E(\alpha)$ for Li_3C_2 monolayer is 60.95 N/m, comparable with silicene (61 N/m)⁷² and germanene (41.4 N/m).⁷²

Since Li atoms are located in the top and bottom surfaces of Li_3C_2 monolayer, it could adsorb hydrogen by strong binding force. We first study the adsorption of a single hydrogen on the Li_3C_2 monolayer ($\text{H}_2@ \text{Li}_3\text{C}_2$). This is modeled by one H_2 adsorbed on the hollow (namely, the H site above the center of the trigon), bridge (namely, the B site above the midline of Li–Li atom), and top (namely, T site above Li atom) sites. These sites have been marked in Figure 1c and the optimized structures of $\text{H}_2@ \text{Li}_3\text{C}_2$ are shown in Figure 3a. The hydrogen adsorption energy E_{ad} is calculated by the following formula:

$$E_{\text{ad}} = \frac{E_{\text{Li}_3\text{C}_2} + nE_{\text{H}_2} - E_{n\text{H}_2@ \text{Li}_3\text{C}_2}}{n} \quad (1)$$

where $E_{n\text{H}_2@ \text{Li}_3\text{C}_2}$, $E_{\text{Li}_3\text{C}_2}$, E_{H_2} , and n are the total energy of $n\text{H}_2@ \text{Li}_3\text{C}_2$, the energy of Li_3C_2 monolayer, the energy of H_2 molecule, and the number of H_2 molecules, respectively. The calculated adsorption energies of single H_2 are 0.121, 0.123, and 0.233 eV/ H_2 at H, B and T sites, respectively (Table 1). Our results indicate that H_2 is more preferable to be adsorbed on T site and have a tendency to tilt toward the Li atom. The bond length of H and H atoms is 0.765 Å, which is slightly longer than that of free H_2 (0.753 Å). Bader analysis demonstrates that Li atoms transfer 0.039 e to H_2 , occupying the antibonding orbital of H_2 . According to molecular orbital theory, the weak stability of H_2 results in the elongation of the H–H bond length.

We next consider the two H_2 molecules adsorbed on the same side of Li_3C_2 monolayer ($2\text{H}_2@ \text{Li}_3\text{C}_2$). The results suggest that the adsorption energies are decreased to 0.08–0.13 eV/ H_2 for three considered cases of T + B, T + H, and B + H, indicating weak attractions between two H_2 molecules and Li_3C_2 monolayer. This is understood that after adsorption, H_2 is negative charged, and coulomb repulsions between H_2 and H_2 leads to the decreased adsorption energy with the increasing number of adsorbed H_2 . Based on the optimal adsorption energy range of 0.2–0.6 eV/ H_2 ,⁶ we conclude that two H_2 molecules cannot be adsorbed on the same side. Therefore, we then explore the double-sided adsorption of hydrogen. Two H_2 molecules for the double-sided adsorption with T + T' configuration indicated in Figure 3c exhibit the largest adsorption energy of 0.223 eV/ H_2 . When the third H_2 molecules is adsorbed, the adsorption energy will be reduced (Figure 3c and Table 1). The hydrogen storage capacity of Li_3C_2 monolayer is calculated to be 8.2 wt %, far exceeding the values of intrinsic 2D materials (Table 2), such as graphene, MoS_2 and $h\text{-BN}$, and comparable to the metal-decorated 2D materials (Table 3), such as $\text{Li}@ \text{graphene}$, $\text{Mg}@ \text{silicene}$, and $\text{Ca}@ \text{borophene}$.

The partial density of states (PDOS) of intrinsic Li_3C_2 monolayer and two H_2 double-sided adsorbed Li_3C_2 monolayer have been calculated to understand the binding

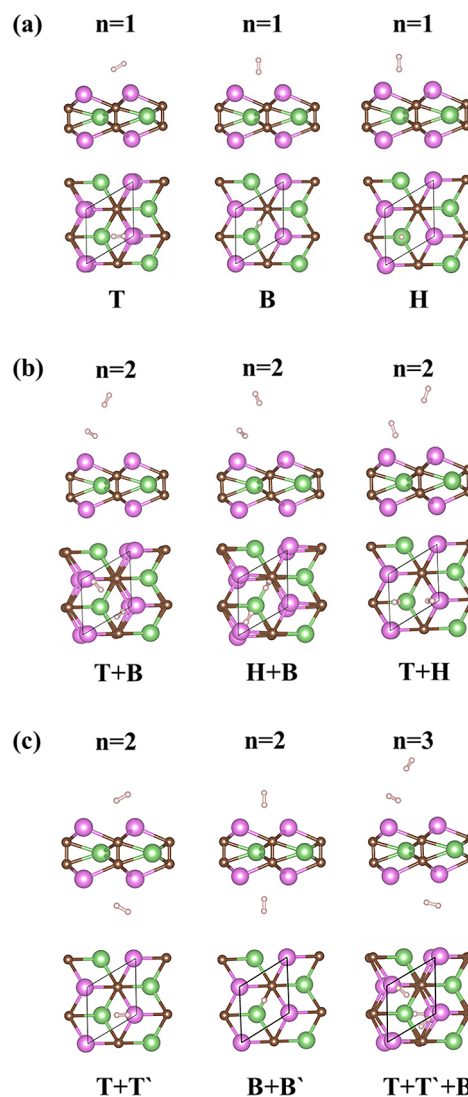


Figure 3. Absorption sites of H_2 molecules on Li_3C_2 monolayer in the side and top views: (a) one H_2 molecule, (b) two H_2 molecules adsorbed on the upper surface, and (c) two or three H_2 molecules adsorbed on the upper and lower surfaces. The symbol n represents the number of adsorbed H_2 molecules. The pink, green, and brown spheres are surface-layer Li, middle-layer Li, and middle-layer C atoms, respectively. H, B, and T represents adsorption sites marked in Figure 1b. T' and B' are mirror-symmetric adsorption sides of T and B, respectively.

mechanism of H_2 molecule to Li_3C_2 . As illustrated in Figure 4a, partial occupation of 2p orbitals of Li can be clearly demonstrated for the Li_3C_2 monolayer. The strong hybridization of C 2p and Li 2p orbitals occurs since C atoms back-donate some of their obtained electrons to Li 2p orbitals. The electron back-donation process usually occurs in quasi-planar hypercoordinate systems⁶⁵ and metal-doped graphene.²⁷ Figure 4b shows that, after the adsorption of H_2 molecules, the Li sp orbitals are hybridized with the H s orbital at −9.8 eV to −8.6 eV, indicating that the hybridization of orbitals plays a key role in the binding of H_2 molecules to Li_3C_2 .

In the onboard application, the hydrogen storage system should be stable under operating conditions, namely, it could adsorb or release hydrogen at room temperature and moderate pressures of 0.2–10 MPa. Therefore, the effect of temperature and pressure on the structural stability has been explored,

Table 1. Hydrogen Adsorption Energies (E_{ad}) at Different Adsorption Sites of the Li_3C_2 Monolayer

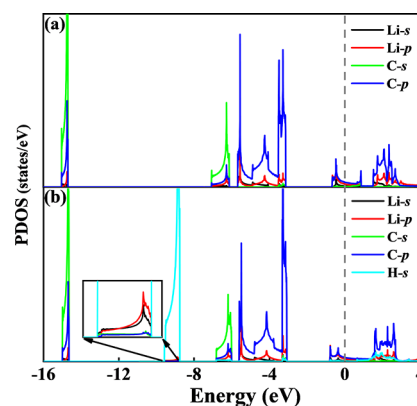
number of adsorbed H_2 molecules, N	site	E_{ad} [eV/ H_2]
1	T	0.233
1	B	0.123
1	H	0.121
2	T + B	0.130
2	H + B	0.128
2	T + H	0.081
2	T + T'	0.223
2	B + B'	0.122
3	T + T' + B	0.163

Table 2. Summary of the Hydrogen Storage Capacities of Carbon Nanostructures and Other Common 2D Materials

	compounds	storage capacity [wt %]	ref
carbon nanostructures	graphene	0.4	7
carbon nanostructures	carbon nanotube	0.4	9
carbon nanostructures	carbon fiber	0.2	12
carbon nanostructures	graphite nanofiber	0.4	8
carbon nanostructures	single-walled carbon nanotube	4.2	10
carbon nanostructures	C_{60}	4.8	11
other 2D materials	MoS_2	<1	15
other 2D materials	h -BN	2.6	16
other 2D materials	silicene	0.9	13
other 2D materials	BCN	5.1	14

Table 3. Summary of the Hydrogen Storage Capacities and Hydrogen Adsorption Energies (E_{ad}) of Carbon Nanostructures and 2D Materials by Metal Decoration

compound	E_{ad} [eV/ H_2]	storage capacity [wt %]	ref
Metal-Decorated Carbon Nanostructures			
Li/Na/K/Ca/Al @graphene	0.17–0.29	6.0–13.8	25,27
Ca@zigzag graphene nanoribbon	0.20	5.1	20
Ti/Li/Ca@ C_{60}	~0.11	2.9–6.2	11, 21, 32
Ti/Li@single-walled carbon nanotube	0.26–0.34	8/2.5	26, 33
Ca@graphdiyne	0.20	17.95	19
Metal-Decorated 2D Materials			
Li/Na/Ca@borophene	0.19–0.25	10.4–14.0	24
Li@ Hf_2CF_2	0.16	10.8	22
Mg@silicene	0.15	8.1	23
Ca@ BC_7	0.26	5.0	28
Ca@ BC_2N	0.30	8.4	17
Li@BN	0.16	7.5	31
Li/Na/K@ BN_2	0.16–0.21	6.6–6.9	30
Li@ BC_3	0.34	10.11	29
K/Ca@germanene	0.09–0.15	12.5–13.9	18

**Figure 4. Partial electronic density of states (PDOS) of (a) Li_3C_2 and (b) $2\text{H}_2@ \text{Li}_3\text{C}_2$. The Fermi energy is set to zero.**

based on the relative energy calculations. The relative energy is defined by the following equation:

$$E_r = E_{n\text{H}_2@ \text{Li}_3\text{C}_2} - E_{\text{Li}_3\text{C}_2} - n[E_{\text{H}_2} + u_{\text{H}_2}(T, P)] \quad (2)$$

where $E_{n\text{H}_2@ \text{Li}_3\text{C}_2}$, $E_{\text{Li}_3\text{C}_2}$, E_{H_2} and n have same meaning as the representatives in eq 1. The term $u_{\text{H}_2}(T, P)$ is the chemical potential of hydrogen at a given temperature T and pressure P , and it can be calculated using the equation

$$u_{\text{H}_2}(T, P) = \Delta H - T\Delta S + k_B T \ln\left(\frac{P}{P^0}\right) \quad (3)$$

where P^0 is the standard atmospheric pressure of 1 bar, k_B the Boltzmann constant, and $\Delta H - T\Delta S$ the change in chemical potential from 0 K to target temperature at ambient pressure. The values of $\Delta H - T\Delta S$ can be obtained from the thermochemical tables,⁷³ and we have listed these values in Table 4.

Table 4. Values of $\Delta H - T\Delta S$ Obtained from the Thermochemical Tables

T (K)	ΔS ($\text{J K}^{-1} \text{mol}^{-1}$)	ΔH (kJ mol^{-1})
0	0	0
100	100.73	3.00
200	119.41	5.69
300	130.86	8.52
400	139.216	11.43

As shown in Figure 5a, under standard atmospheric pressure, $\text{H}_2@ \text{Li}_3\text{C}_2$ and $2\text{H}_2@ \text{Li}_3\text{C}_2$ structures have negative relative energies at 0 K, implying that they are stable. As the temperature increases, the relative energies are gradually increased. When temperature exceeds the critical temperature at which the relative energy is zero, the structure is no longer stable and undergo spontaneous hydrogen desorption. The critical temperatures for $\text{H}_2@ \text{Li}_3\text{C}_2$ and $2\text{H}_2@ \text{Li}_3\text{C}_2$ structures are 239 and 226 K, respectively. Figure 5b suggests that after considering the pressure, the critical temperature could be increased. Hydrogen storage in Li_3C_2 monolayer remains stable at room temperature with pressures up to 2.7 MPa for $\text{H}_2@ \text{Li}_3\text{C}_2$ and 4 MPa for $2\text{H}_2@ \text{Li}_3\text{C}_2$, which are in the moderate pressure range.

The van't Hoff equation⁷⁴ has been used to estimate the hydrogen desorption temperature (T_D) in the range of equilibrium pressures (1–10 atm), which is defined as

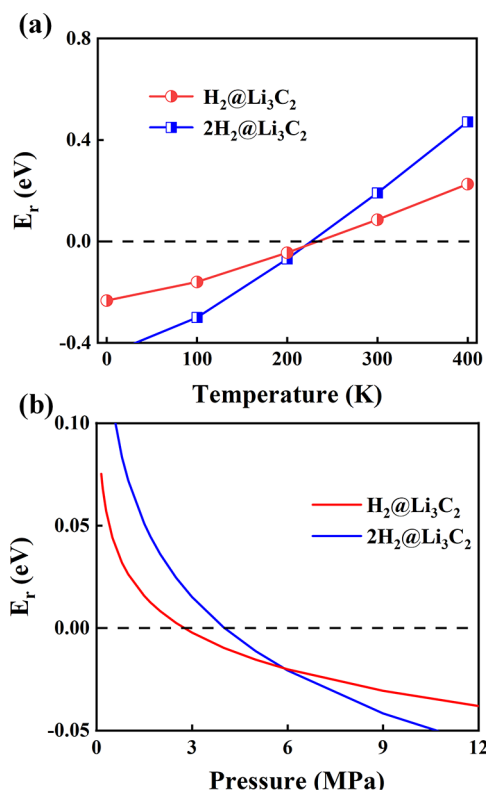


Figure 5. (a) Relative energies of H_2 adsorbed on Li_3C_2 monolayer, as the function of temperature, under standard atmospheric pressure ($P = 0.1$ MPa). (b) The relative energies of H_2 adsorbed on Li_3C_2 monolayer as a function of pressure at 300 K.

$$T_D = \frac{E_{ad}}{k_B[(\Delta S/R) - \ln p]} \quad (4)$$

where E_{ad} is the hydrogen adsorption energy, k_B the Boltzmann constant, ΔS the change of entropy from molecular gas to dissolved solid hydrogen (which is approximately the standard hydrogen entropy, 130 J/K mol), R the gas constant ($R = 8.31$ J/(K mol)), and p the equilibrium pressure. Using the average E_{ad} of 0.223 eV/ H_2 , we can estimate the average T_D . By using the E_{ad} value of the second H_2 absorbed by Li_3C_2 monolayer (0.213 eV/ H_2), we can deduce the onset of the desorption at the minimum T_D . Similarly, by using the E_{ad} value of the first H_2 absorbed by Li_3C_2 monolayer (0.233 eV/ H_2), we can also deduce the maximum T_D required for a full discharge of hydrogen from the medium. In Figure 6, we present the estimated minimum, average, and maximum T_D s, as a functional of pressure. Results illustrate that the minimum, average, and maximum T_D values at normal atmospheric pressure are 272, 286, and 298 K, respectively, which could be further increased by increasing the equilibrium pressure.

In order to further check the T_D evaluations, we performed AIMD calculations for $4 \times 4 \times 1$ $2H_2@Li_3C_2$, including 36 H_2 molecules at 200 and 300 K (Figure 7). According to the calculated hydrogen desorption temperature at ambient pressure, hydrogen should remain adsorbed at 200 K and hydrogen desorption could occur at 300 K. As shown in Figure 7a, indeed, we can see that only 6.25% of the hydrogen molecules break free from the surface after a 6 ps run at 200 K (hydrogen molecules spaced more than 2.5 Å apart from the surface are viewed as undergoing desorption). The hydrogen storage capacity is as high as 7.69 wt %, which is larger than

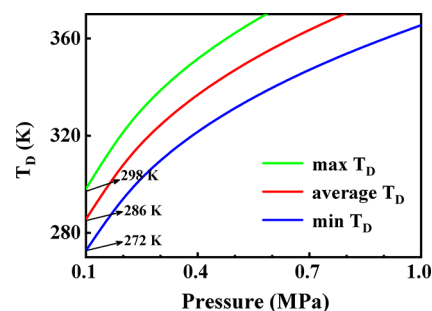


Figure 6. Dependence of hydrogen desorption temperature (T_D) on the equilibrium pressure. The green, red, and blue lines represent the maximum, average, and minimum T_D values, respectively.

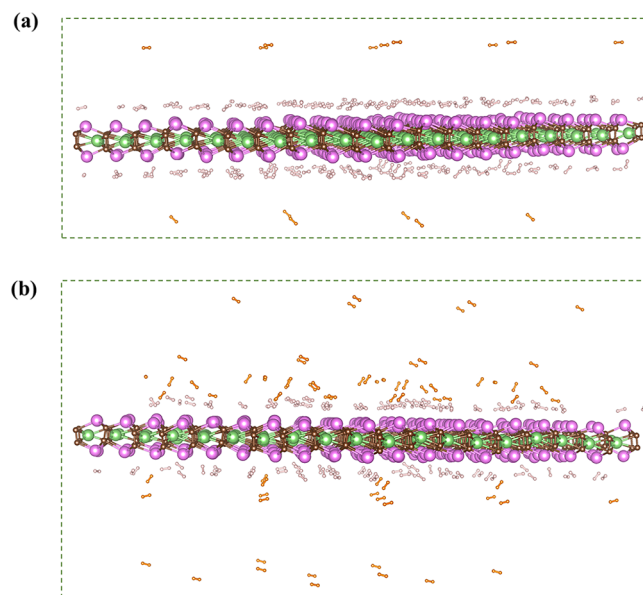


Figure 7. Structure images of $2H_2@Li_3C_2$ at (a) 200 K and (b) 300 K after 6 ps simulations. The desorbed H_2 molecules are represented by orange spheres.

that of some metal-decorated 2D materials. As the temperature increases to 300 K, 37.5% of the hydrogen molecules move away from the Li_3C_2 surface (Figure 7b), indicating the effectiveness of hydrogen storage/release cycles.

These findings demonstrate that Li_3C_2 monolayer is indeed a promising choice for efficient hydrogen storage media. Therefore, remarks on the possible synthesis of Li_3C_2 monolayer are necessary. We selected Cu and graphene as the substrates for the growth of Li_3C_2 monolayer, since they are widely used as substrates in experiments. In addition, Cu and graphene have very similar lattice parameters as Li_3C_2 monolayer, namely, the lattice mismatches are 4.3% and 3.9% for the $Li_3C_2/(4 \times 4)$ -Cu(111) and Li_3C_2 /graphene, respectively.

The cohesive energies between Li_3C_2 monolayer and the substrates of Cu and graphene have been calculated using the following equation:

$$E_{coh} = \frac{E_{slab_Cu} + E_{Li_3C_2} - E_{Li_3C_2/Cu}}{N} \quad (5)$$

$$E_{coh} = \frac{E_{graphene} + E_{Li_3C_2} - E_{Li_3C_2/graphene}}{A} \quad (6)$$

where $E_{\text{Li}_3\text{C}_2/\text{Cu}}$, $E_{\text{slab_Cu}}$, $E_{\text{Li}_3\text{C}_2}$, $E_{\text{Li}_3\text{C}_2/\text{graphene}}$, and E_{graphene} are the energies of the $\text{Li}_3\text{C}_2/\text{Cu}(111)$, $\text{Cu}(111)$ substrate, Li_3C_2 monolayer, $\text{Li}_3\text{C}_2/\text{graphene}$, and graphene, respectively, and N and A are the number of atoms of Li_3C_2 monolayer and the surface area of graphene, respectively. After structural optimization, no strong chemical bonds are formed between Li_3C_2 monolayer and the two substrates (Figure 8). The

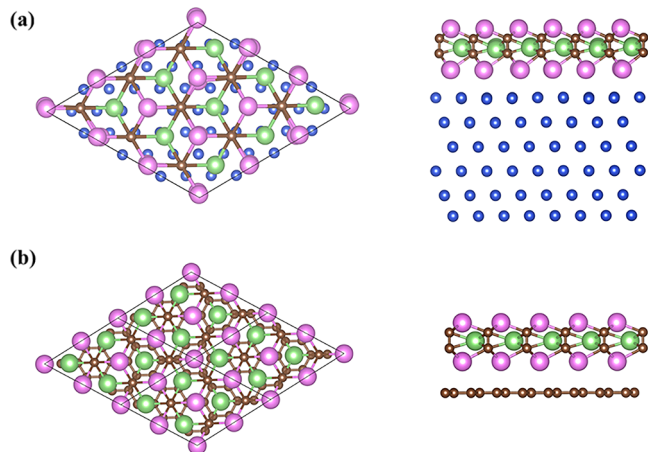


Figure 8. Top and side views of the optimized structures for (a) $\text{Li}_3\text{C}_2/(4 \times 4)\text{-Cu}(111)$ and (b) $\text{Li}_3\text{C}_2/\text{graphene}$, respectively. The pink, green, blue and brown spheres are surface-layer Li, middle-layer Li, Cu, and C atoms, respectively.

cohesive energy is 0.13 eV/atom between Li_3C_2 monolayer and $\text{Cu}(111)$ substrate, comparable to that (0.3 eV/atom) of β_{12} boron sheet on Ag substrate.⁷⁵ The cohesive energy of $\text{Li}_3\text{C}_2/\text{graphene}$ is 10.85 meV/ $\text{\AA}^2$, which is comparable to that of bilayer BN/graphene (8.56 meV/ $\text{\AA}^2$).⁷⁶ The results indicate the possibility to grow Li_3C_2 monolayer on $\text{Cu}(111)$ surface and graphene using molecular beam epitaxy.

Considering the superconductivity of LiC_6 monolayer, we have explored the superconducting critical temperature (T_c) of Li_3C_2 monolayer, which is estimated based on the Allen–Dynes-modified McMillan equation⁷⁷ with a typical choice of Coulomb pseudopotential of $\mu^* = 0.1$. Calculated results show the negligible superconductivity of Li_3C_2 monolayer, with a T_c of 0.1 K. The T_c value of monolayer Li_3C_2 (8.1 K) is smaller than that of monolayer LiC_6 , which originated from the smaller EPC parameter ($\lambda = 0.25$) in Li_3C_2 than that of LiC_6 ($\lambda = 0.61$).⁵⁹

In summary, we predict a thermodynamically stable Li_3C_2 monolayer using the CALYPSO method and first-principles calculations, which is a promising hydrogen storage medium without metal decoration. Interestingly, carbon atoms in Li_3C_2 form a phC moiety with six neighboring Li atoms and one C atom. The hydrogen adsorption energy of Li_3C_2 monolayer is 0.22 eV/ H_2 , and the storage capacity is as high as 8.2 wt %, far exceeding that of most intrinsic 2D materials. Further analysis has demonstrated that the mechanism for binding the attached H_2 molecules mainly comes from the hybridization of H s and Li sp orbitals. Significantly, for the Li_3C_2 monolayer, the realization of hydrogen storage can be achieved at room temperature and at pressure of 40 bar. Hydrogen desorption can occur at room temperature without external pressure, based on the van’t Hoff equation. AIMD simulations demonstrate the thermal stability of Li_3C_2 after hydrogen adsorption and the effectiveness of hydrogen charge/release.

These results indicate that Li_3C_2 is a potential hydrogen storage medium for onboard hydrogen storage equipment to be used in light-duty vehicles. We demonstrate that the $\text{Cu}(111)$ surface and graphene are suitable substrates for the growth of the Li_3C_2 monolayer. The current results provide an effective guide to design intrinsic 2D hydrogen storage materials with high storage capacity and desorption temperature.

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Author Contributions

J.C. performed the data curation and formal analysis; S.L. was responsible for investigation and methodology; M.X. performed the conceptualization and writing (original draft, as well as review and editing); F.W. and Y.S. were involved with the investigation; J.H. participated in the writing (original draft, as well as review and editing); and Y.L. was responsible for funding acquisition and writing (review and editing). CRediT: **Jingyan Chen** data curation, formal analysis; **Shuyi Lin** investigation, methodology; **Meiling Xu** conceptualization,

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Notes

The authors declare no competing financial interest.

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